

# Transport Layer

Module - III

Time - 2:33:04

Transport layer is lies over inside the devices.

transportation of the data from the network to the application that is done by transport layer.

\* From one Network to another network the data transfer by Network layer  
(The actual transportation from one computer to another computer.)

\* But within that computer transportation of the data from Network  $\rightarrow$  Application by Transport layer

Application

My whatsapp.

My computer

His computer

His whatsapp

Application

Transport layer

Network layer

Transport layer

Checksums: It is a error-detection mechanism used by TCP/UDP to check data has been corrupted or not during transmission

My computer

data.

checksum  
100

data  
checksum

His computer

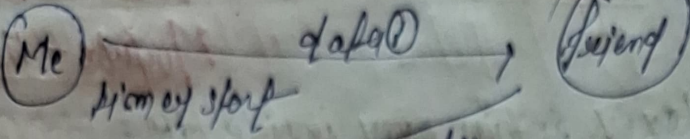
data  
checksum

100

90 x error.

data corrupted

## Timers.



timer stop ← i got the data

if timer expires then the data not got by friend.

## UDP (User Datagram Protocol)

→ data may or may not be delivered.

→ data may change / may not be in order.

It is a connectionless protocol.

UDP uses checksum → if data corrupted, OK  
no problem

## UDP Packets.

|                                      |                                    |                            |
|--------------------------------------|------------------------------------|----------------------------|
| Source Port Number<br>(2 bytes)      | length of<br>checksum<br>(2 bytes) | } Header<br><u>8 bytes</u> |
| Destination Port Number<br>(2 bytes) | checksum.<br>(2 bytes)             |                            |
| data.                                |                                    |                            |

∴ total size is 16

$2^{16} - 8 \approx 65,536$  bytes → size of the data we can send in a packet.

## Uses.

- It's very fast
- video conf app.
- DNS → UDP
- Gaming.

# # TCP - Transmission control protocol

3:3:14

- Transport layer protocol.
- Application layer send lot of raw data then TCP segments that data.
  - ↳ divide in chunks.
  - ↳ add Header
  - ↳ check sums. all that.
- congestion control.
- Maintain the order of data if data not received. it take care of that.

## Features.

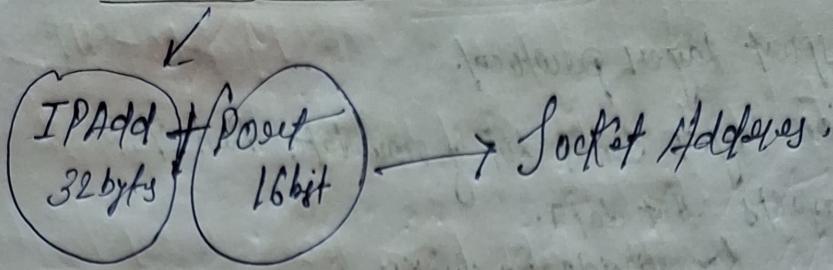
- connection oriented.
- Error control
- congestion control.
- full duplex.

## 3-Way Handshake

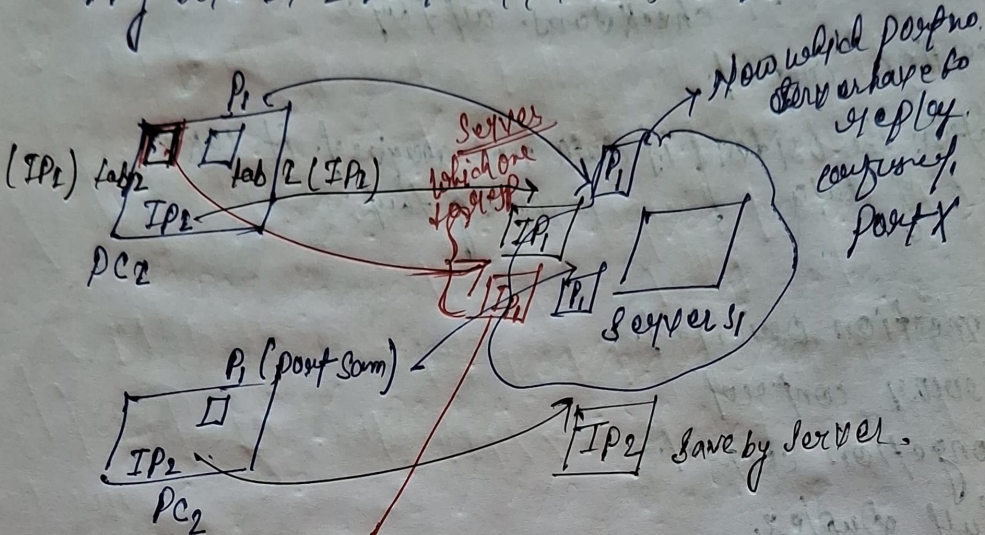
go through the vid  
cant write! 3:10:29

## Socket Address.

Gate Semesters L-68.



Why both IP & Port address is used?



### IP Addr.

In same computer if another tab sent the request then the IP Addr of Tab 2  $\neq$  Tab 1. In this case which tab to respond by the server is not simple.

So only IP Addr can't be used.

### Port Number.

From two diff computer, there can be possible that Application have the same port No. In both two computer in that case which computer to give the response, it confusing. from server side.

$\therefore$  This is the two key reason why we use Socket Address. Not use only port or IP.

# TCP connection Establishment (3-way handshake)

Client (Machine 1)

Server (Machine 2)

It is for connection establish

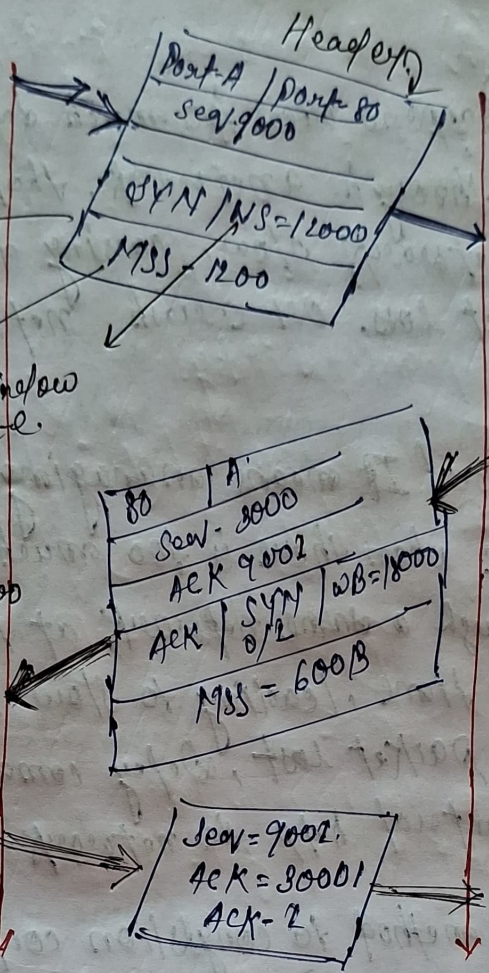
maximum segment size

Client can only accept buffer up to 12000

At a time a single packet size should be 1200 only.

Time

Time



TCP (Transmission control protocol)

UDP (User Datagram protocol)

It is a connection oriented

① connectionless

- ② Reliable (order follow)
- ③ Error control mandatory.
- ④ Slow Transmission.
- ⑤ More overhead
- ⑥ Flow control
- ⑦ congestion control

- ② less Reliable (no order)
- ③ Error control is optional.
- ④ Fast Transmission.
- ⑤ less overhead.
- ⑥ No flow control
- ⑦ No congestion control.

## Congestion control:

Congestion happens when too much data goes through a network, that makes network slow, causing delay, packet loss, or crash network.



Congestion control It means managing network traffic when there is too much data trying to pass through a same network at the same time, leading to slow internet speed, packet loss, delay communication. Congestion control help the network to work properly. There are two method to congestion control

### (i) Open loop congestion control (Prevention method)

before it occurs. This control/prevent congestion by using proper design rules. It is handled by the sender & receiver.

(i) Retransmission policy: The sender will resend the lost or corrupted packet carefully to avoid congestion.

(ii) Window policy: Control how many packets are sent at once.

(iii) Acknowledgment policy: Instead of ack every time for each packet, the receiver consent Ack in group together.

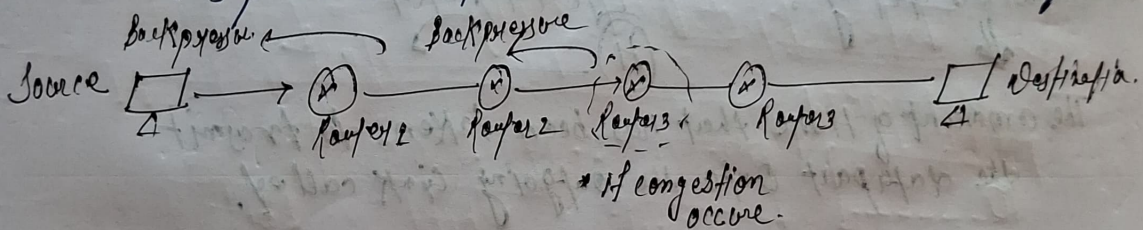
(iv) Dropping policy: Router will discard less important packet during congestion to maintain network flow.

(v) Admission policy: Only a limited number of users can send the data at a time.

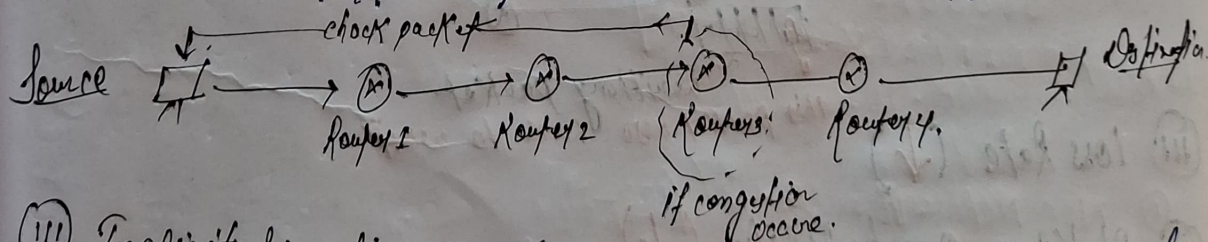
(vi) Closed loop congestion control (after congestion happens) <sup>Handling.</sup>  
When congestion is detected, that system slow down data transmission until the congestion is cleared. These techniques ensure smooth data flow in network.

Closed loop congestion control, detect and control congestion by using feedback mechanism (feedback from Network)

(i) Backpressure: If a router is congested, then it inform the previous router to slow down data transmission this process continues back to back to the source/client



(ii) Choke packet: A Special packet sent by the router to the sender (Source) to reduce the data flow.



(iii) Implicit Signaling: The source / or the computer only identify the congestion indirectly by observing delays or lost packet.

(iv) Explicit Signaling: The router directly inform the computer about the congestion

(i) Backward signaling. (ii) Forward signaling.

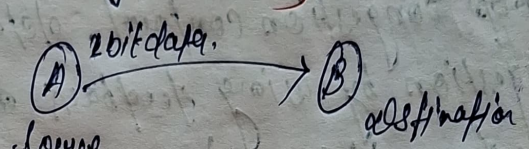
# QoS (Quality of service)

5 minit engineering

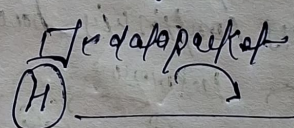
① Bandwidth (↑) the max amount of data that can be transmitted over a network in given time period.

→ Bandwidth should be high.

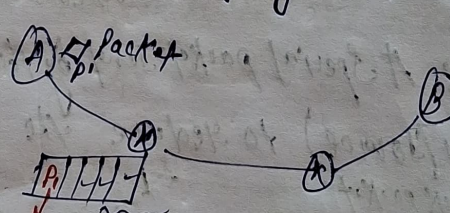
② Delay (↓)   
 → Propagation delay   
 → Transmission delay.   
 → Queue delay.   
 } Delay should be low

Propagation delay:   


The amount of time that has been taken for the bit to propagate from A to B call...   
 More distance more delay.

Transmission delay:   


The amount of time that has been taken to transmit the datapacket onto the outgoing link call ed.

Queue delay:   


waiting packet   
 this packet also have to wait.

③ Loss Rate (↓)

④ Blocking Probability (↓)

⑤ Delay jitter (↓) different packets have different delay so, the diff of max delay and min delay Jitter.

